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Course code: DSA0607

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Course name: Data handling& visualization

Slot D

11.

Code:

```
# Geographic Data
```

```
City <- c("City A",  
          "City B",  
          "City C")
```

```
Population <- c(500000,  
                700000,  
                600000)
```

```
Avg_Temperature <- c(75,  
                     68,  
                     80)
```

```
Elevation <- c(1000,  
              800,  
              1200)
```

```
geo <- data.frame(City,  
                  Population,  
                  Avg_Temperature,  
                  Elevation)
```

```
print(geo)
```

```
plot(1:3,  
     1:3,  
     pch=19,  
     col="blue",  
     xaxt="n",  
     yaxt="n",  
     xlab="",
```

```

ylab="",
main="Geographic Distribution of Cities")

text(1:3,
     1:3,
     labels=geo$City,
     pos=3)

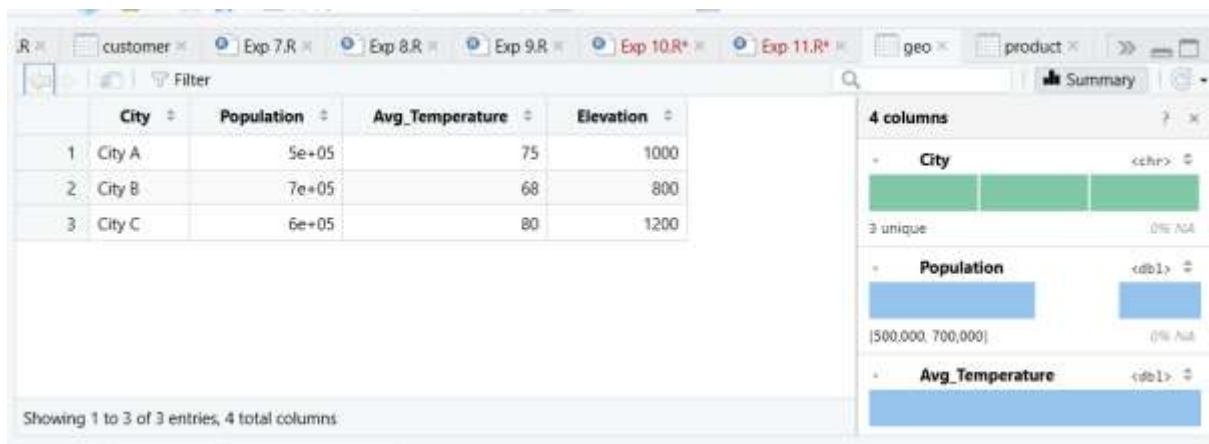
plot(geo$Avg_Temperature,
     geo$Population,
     pch=19,
     col="red",
     xlab="Average Temperature",
     ylab="Population",
     main="Temperature vs Population")

abline(lm(Population ~ Avg_Temperature,
          data=geo),
       col="blue")

```

View(geo)

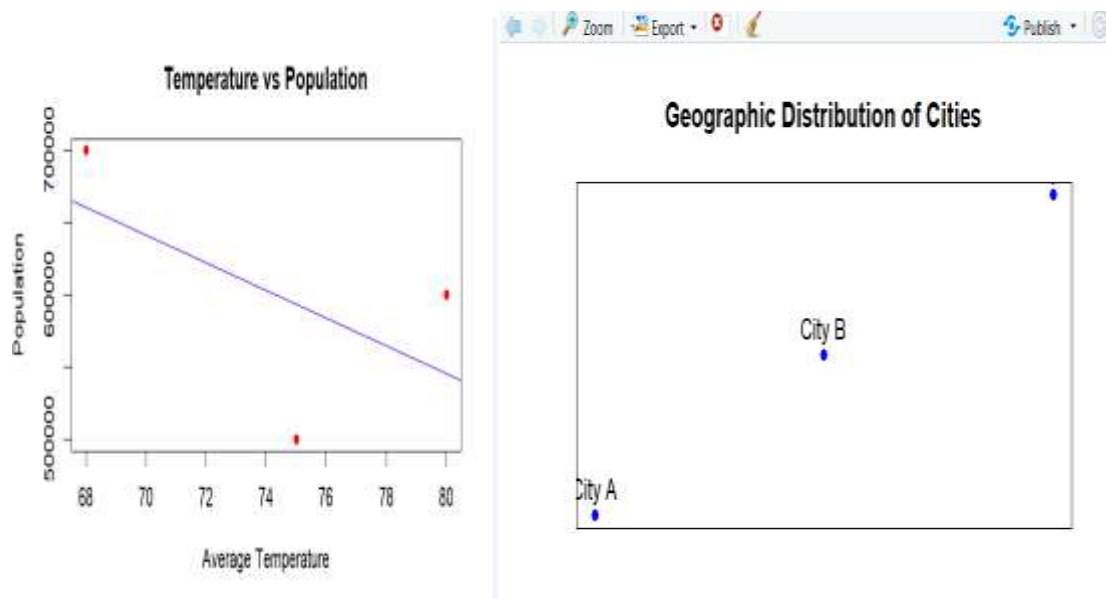
OUTPUT:



	City	Population	Avg_Temperature	Elevation
1	City A	5e+05	75	1000
2	City B	7e+05	68	800
3	City C	6e+05	80	1200

4 columns
City <chr> 3 unique 0% NA
Population <dbl> [500,000, 700,000] 0% NA
Avg_Temperature <dbl>

Showing 1 to 3 of 3 entries, 4 total columns



12.

Code:

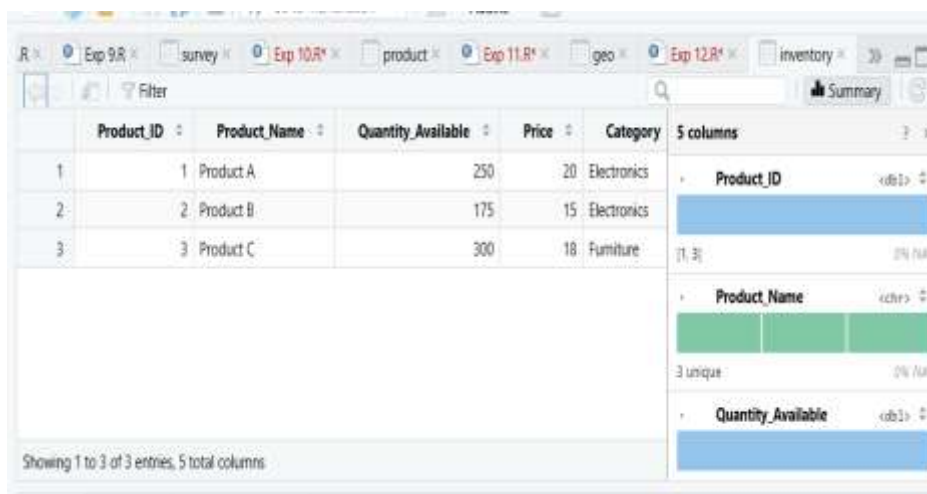
```
# Product Inventory Dataset
Product_ID <- c(1,2,3)
Product_Name <- c("Product A",
                  "Product B",
                  "Product C")
Quantity_Available <- c(250,175,300)
Price <- c(20,15,18)
inventory <- data.frame(Product_ID,
                        Product_Name,
                        Quantity_Available,
                        Price)
print(inventory)
barplot(inventory$Quantity_Available,
        names.arg=inventory$Product_Name,
        main="Quantity Available for Each Product",
        xlab="Product Name",
        ylab="Quantity Available",
        col="skyblue",
```

```

border="black")
inventory$Category <- c("Electronics",
                        "Electronics",
                        "Furniture")
inventory_table <- table(inventory$Category,
                        inventory$Product_Name)
barplot(inventory_table,
        main="Quantity by Product Category",
        xlab="Category",
        ylab="Number of Products",
        col=rainbow(nrow(inventory_table)),
        legend.text=rownames(inventory_table))
View(inventory)

```

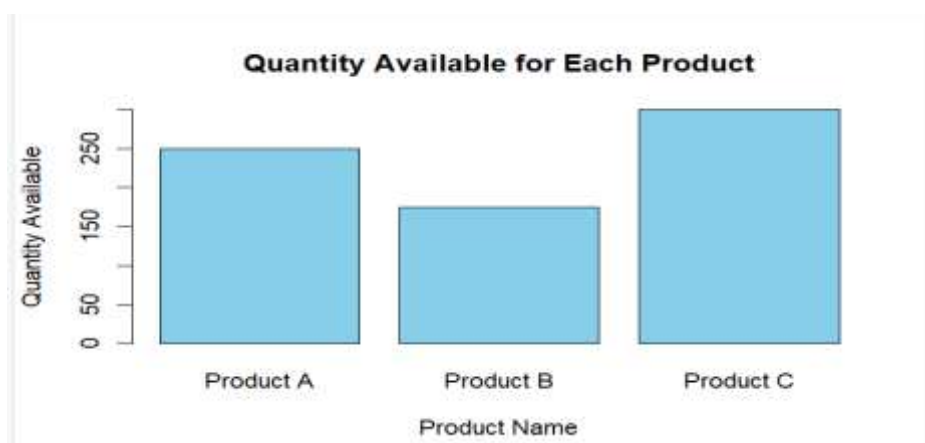
OUTPUT:

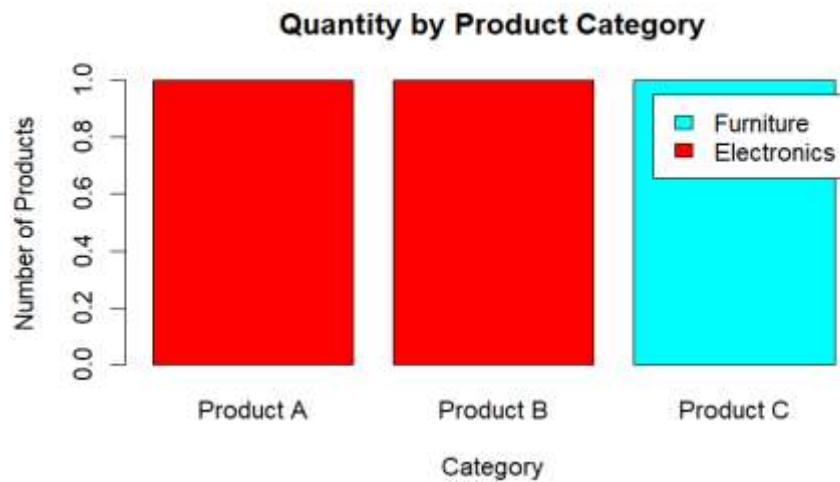


	Product_ID	Product_Name	Quantity_Available	Price	Category
1	1	Product A	250	20	Electronics
2	2	Product B	175	15	Electronics
3	3	Product C	300	18	Furniture

Summary Panel:

- Product_ID**: 5 columns, 3 unique values
- Product_Name**: 3 unique values
- Quantity_Available**: 3 unique values





13.

Code:

```
# Stock Prices Dataset
Date <- c("2023-01-01",
          "2023-01-02",
          "2023-01-03")
Stock_A <- c(100,105,110)
Stock_B <- c(150,152,148)
Stock_C <- c(120,118,122)
stock <- data.frame(Date,
                    Stock_A,
                    Stock_B,
                    Stock_C)

print(stock)

plot(stock$Stock_A,
     type="o",
     col="blue",
     xaxt="n",
     ylim=c(90,160),
     xlab="Date",
     ylab="Stock Price",
```

```

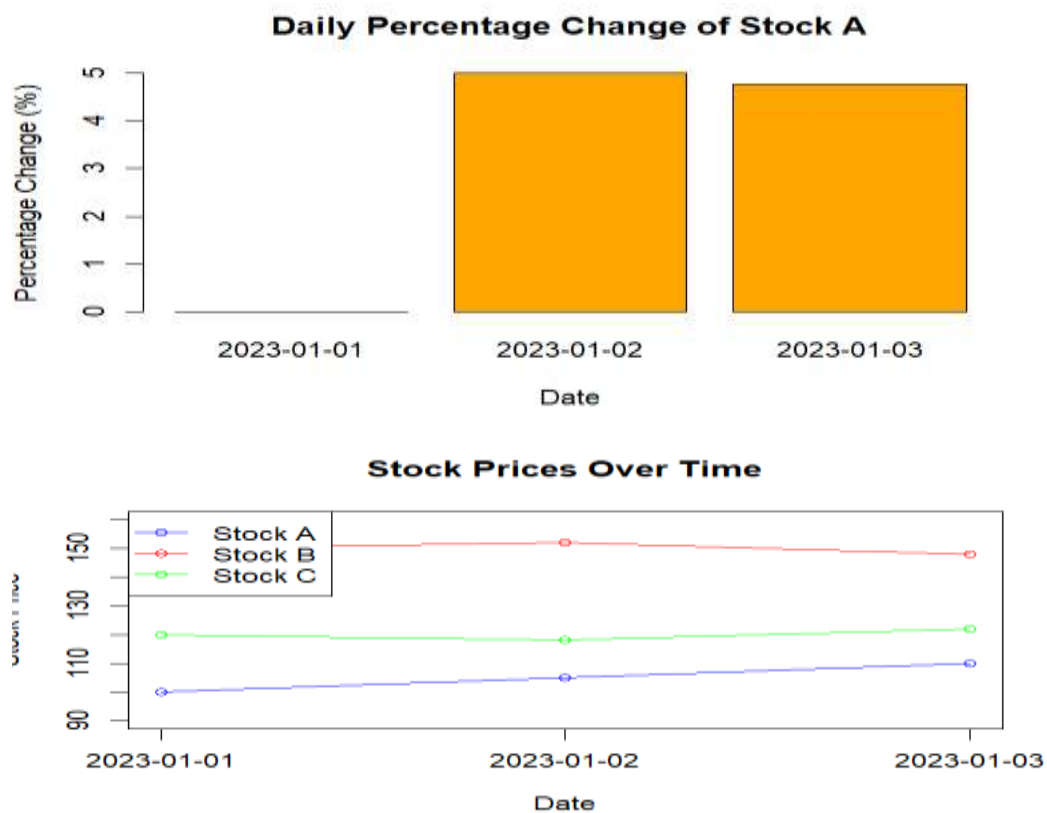
    main="Stock Prices Over Time")
lines(stock$Stock_B,
      type="o",
      col="red")
lines(stock$Stock_C,
      type="o",
      col="green")
axis(1,
     at=1:3,
     labels=stock$Date)
legend("topleft",
      legend=c("Stock A", "Stock B", "Stock C"),
      col=c("blue", "red", "green"),
      lty=1,
      pch=1)
percent_change <- c(0,
                    ((105-100)/100)*100,
                    ((110-105)/105)*100)
barplot(percent_change,
      names.arg=stock$Date,
      main="Daily Percentage Change of Stock A",
      xlab="Date",
      ylab="Percentage Change (%)",
      col="orange",
      border="black")
View(stock)

```

OUTPUT:

	Date	Stock_A	Stock_B	Stock_C
1	2023-01-01	100	150	120
2	2023-01-02	105	152	118
3	2023-01-03	110	148	122

Showing 1 to 3 of 3 entries, 4 total columns



14.

Code:

```
# Energy Consumption Dataset  
Sector <- c("Residential","Commercial","Industrial",  
           "Residential","Commercial","Industrial")  
Region <- c("North","South","West",  
            "East","North","South")  
Month <- c("Jan","Jan","Feb","Feb","Mar","Mar")
```

```

Temperature <- c(15,24,20,18,28,30)
Units_Consumed <- c(320,540,880,350,610,920)
Cost <- c(2100,3600,5900,2300,4100,6200)
Renewable_Usage <- c(22,18,12,25,20,15)
Peak_Hours <- c(4,6,8,5,7,9)
energy <- data.frame(
  Sector,
  Region,
  Month,
  Temperature,
  Units_Consumed,
  Cost,
  Renewable_Usage,
  Peak_Hours
)
print(energy)
hist(energy$Units_Consumed,
  main="Distribution of Units Consumed",
  xlab="Units Consumed (kWh)",
  ylab="Frequency",
  col="skyblue",
  border="black")
plot(density(energy$Units_Consumed),
  main="Density Plot of Units Consumed",
  xlab="Units Consumed",
  col="red",
  lwd=2)
symbols(energy$Temperature,
  energy$Units_Consumed,
  circles=energy$Peak_Hours,

```

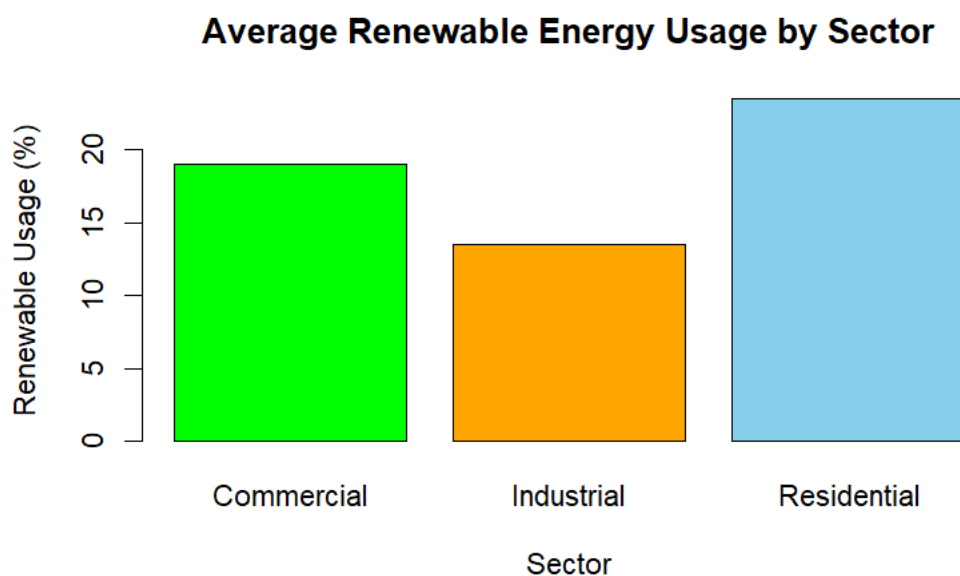


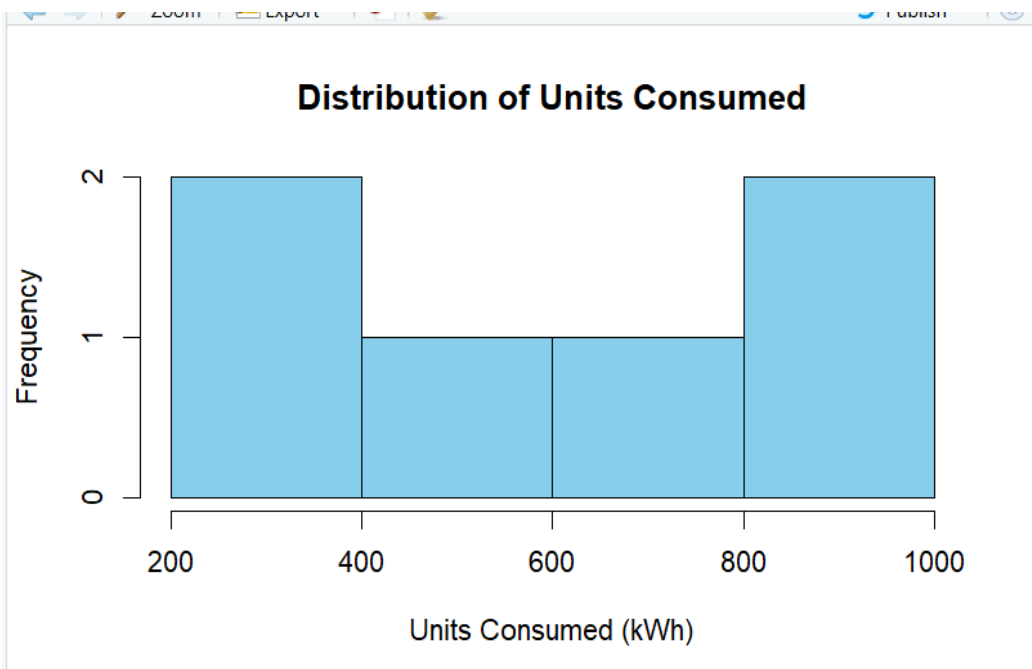
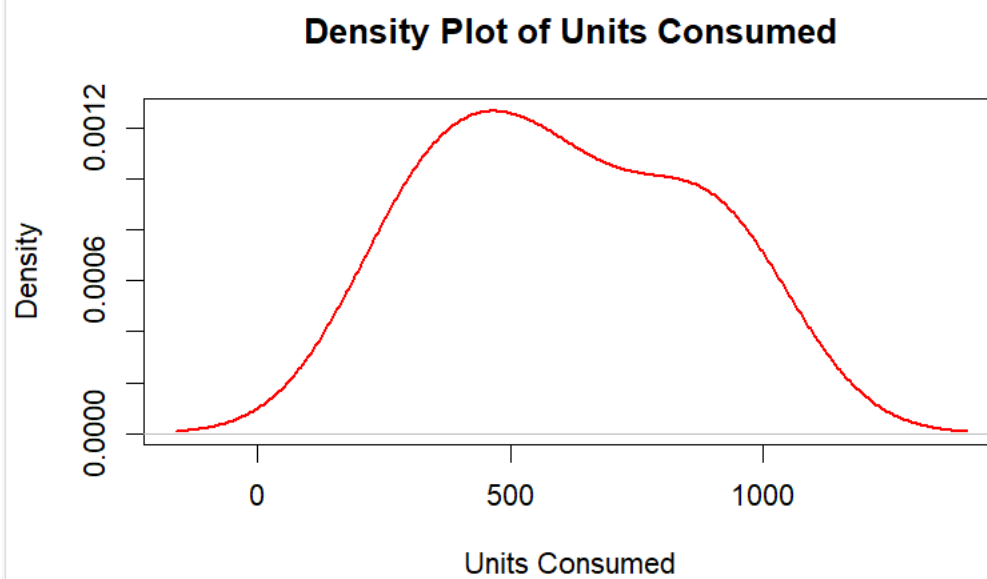
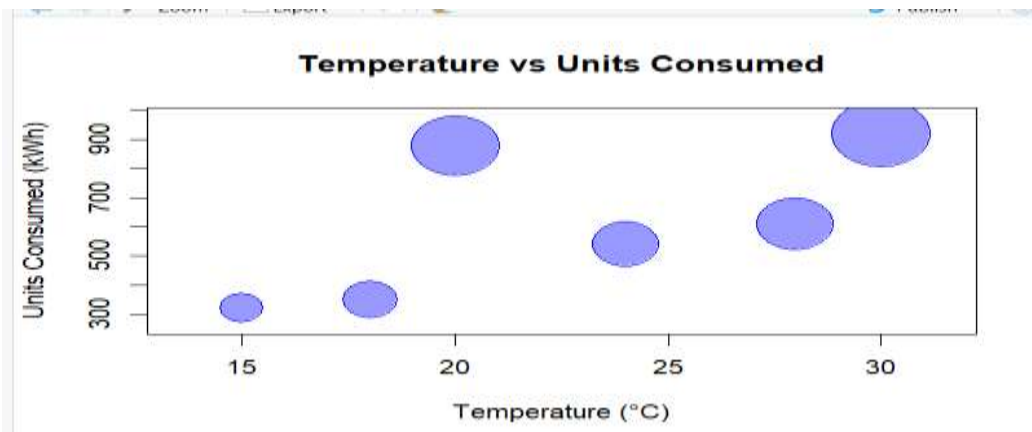
```

    inches=0.3,
    bg=rgb(0,0,1,0.4),
    fg="blue",
    xlab="Temperature (°C)",
    ylab="Units Consumed (kWh)",
    main="Temperature vs Units Consumed")
avg_usage <- aggregate(Renewable_Usage ~ Sector,
                        data=energy,
                        mean)
print(avg_usage)
barplot(avg_usage$Renewable_Usage,
        names.arg=avg_usage$Sector,
        main="Average Renewable Energy Usage by Sector",
        xlab="Sector",
        ylab="Renewable Usage (%)",
        col=c("green","orange","skyblue"))

```

OUTPUT:





15.

Code:

```
# Student Mini Data

Student_ID <- c("S01","S02","S03","S04","S05","S06")
Gender <- c("Male","Female","Male","Female","Male","Female")
Study_Hours <- c(2.0,3.5,1.5,4.0,2.8,3.0)
Attendance <- c(78,90,70,95,85,92)
Math_Score <- c(62,80,55,90,72,82)
Science_Score <- c(65,85,58,92,74,86)
Exam_Date <- c("2025-01-10","2025-01-10",
               "2025-02-12","2025-02-12",
               "2025-03-15","2025-03-15")

student <- data.frame(Student_ID,
                      Gender,
                      Study_Hours,
                      Attendance,
                      Math_Score,
                      Science_Score,
                      Exam_Date)

print(student)

hist(student$Math_Score,
     main="Distribution of Math Scores",
     xlab="Math Score",
     ylab="Frequency",
     col="skyblue",
     border="black")

plot(student$Study_Hours,
     student$Math_Score,
     col=ifelse(student$Gender=="Male","blue","red"),
```

```

pch=19,
xlab="Study Hours",
ylab="Math Score",
main="Study Hours vs Math Score")

abline(lm(Math_Score ~ Study_Hours,
          data=student),
       col="darkgreen",
       lwd=2)

legend("topleft",
       legend=c("Male","Female"),
       col=c("blue","red"),
       pch=19)

student$Exam_Date <- as.Date(student$Exam_Date)
student$Month <- format(student$Exam_Date,"%Y-%m")

avg_math <- aggregate(Math_Score ~ Month,
                      data=student,
                      mean)

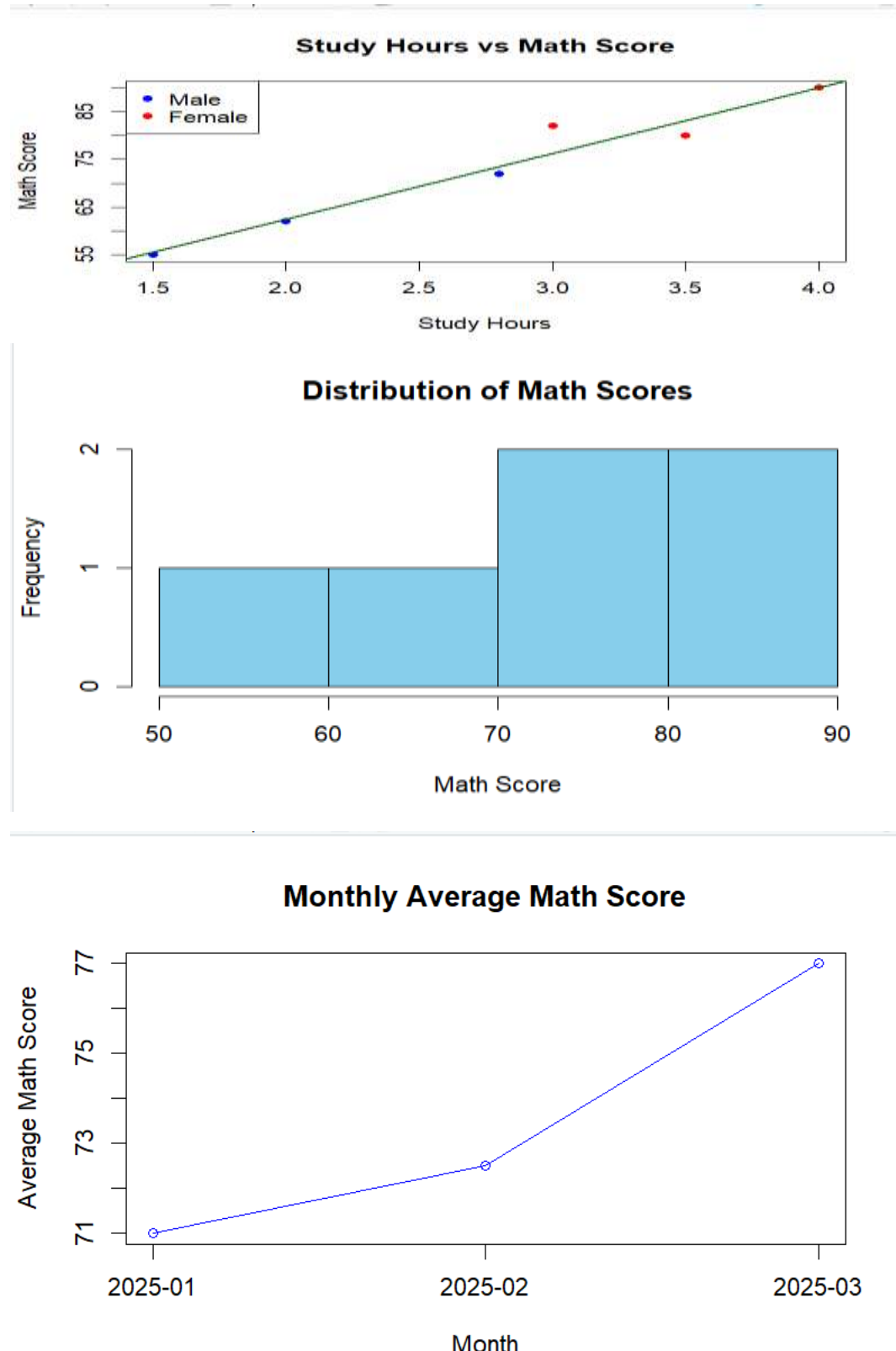
print(avg_math)

plot(avg_math$Math_Score,
     type="o",
     xaxt="n",
     col="blue",
     xlab="Month",
     ylab="Average Math Score",
     main="Monthly Average Math Score")

```

```
axis(1,  
      at=1:nrow(avg_math),  
      labels=avg_math$Month)
```

OUTPUT:



16.

Code: # Patient Health Risk Dataset

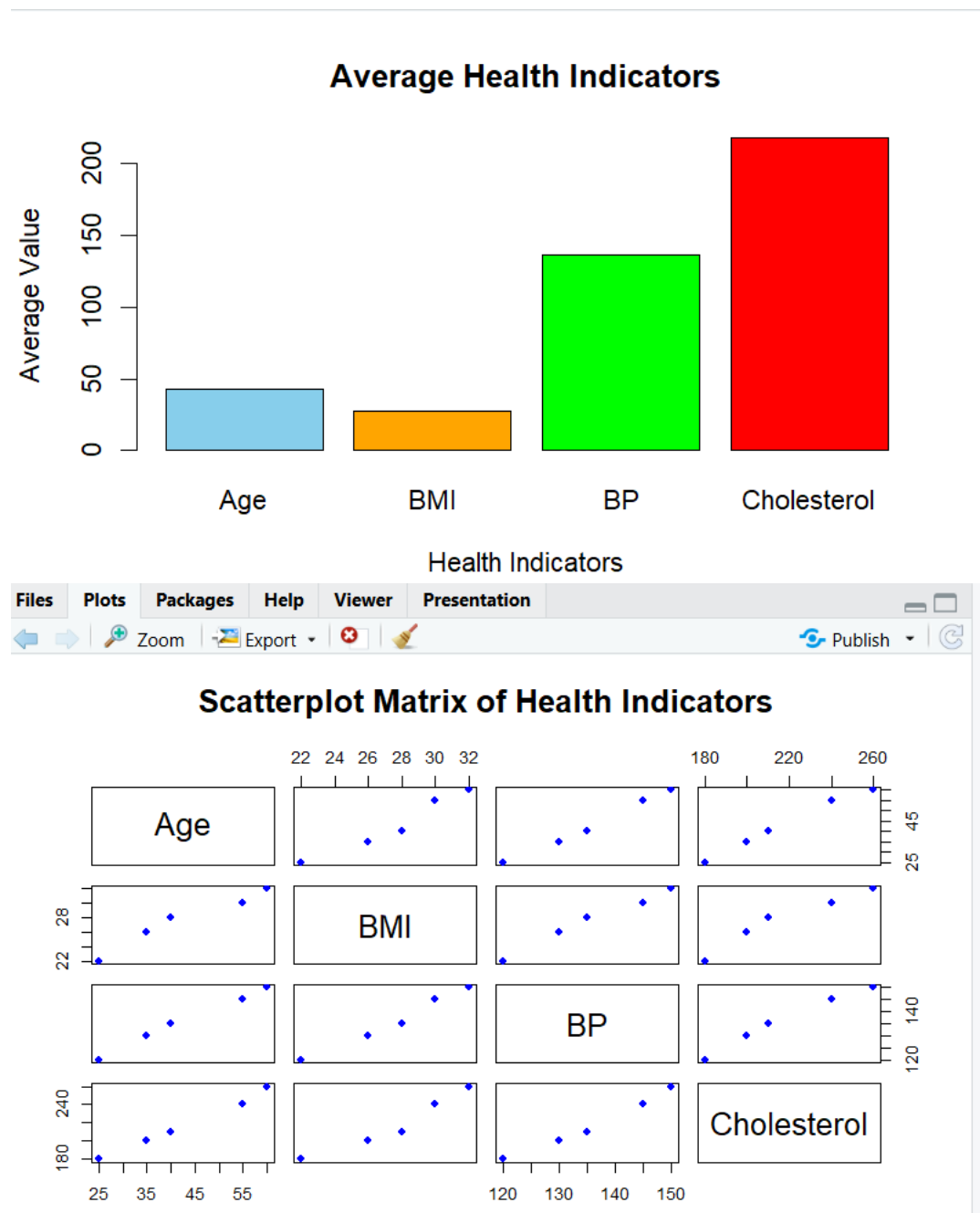
```
Patient_ID <- c("P1","P2","P3","P4","P5")
Age <- c(25,40,55,35,60)
BMI <- c(22,28,30,26,32)
BP <- c(120,135,145,130,150)
Cholesterol <- c(180,210,240,200,260)
patient <- data.frame(
  Patient_ID,
  Age,
  BMI,
  BP,
  Cholesterol
)
print(patient)
pairs(patient[,2:5],
  main="Scatterplot Matrix of Health Indicators",
  col="blue",
  pch=19)
plot(ecdf(patient$Cholesterol),
  main="ECDF of Cholesterol Levels",
  xlab="Cholesterol",
  ylab="ECDF",
  col="blue")
avg_values <- c(
  mean(patient$Age),
  mean(patient$BMI),
  mean(patient$BP),
  mean(patient$Cholesterol)
)
```

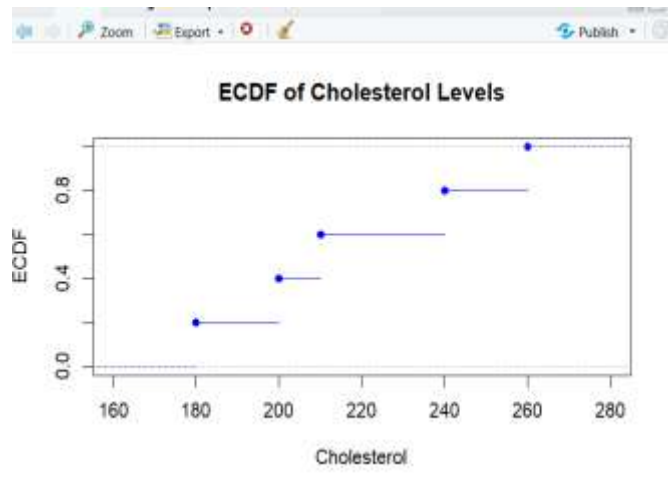
```

barplot(avg_values,
       names.arg=c("Age","BMI","BP","Cholesterol"),
       col=c("skyblue","orange","green","red"),
       main="Average Health Indicators",
       xlab="Health Indicators",
       ylab="Average Value")

```

OUTPUT:





17.

Code:

```
# Vehicle Performance Dataset
Vehicle_ID <- c("V1","V2","V3","V4","V5")
Engine_Size <- c(1.5,2.0,3.0,2.5,1.8)
Horsepower <- c(110,150,250,200,130)
Fuel_Efficiency <- c(18,15,12,14,17)
Top_Speed <- c(180,200,250,220,190)
Safety_Rating <- c(4,5,5,4,3)
vehicle <- data.frame(
  Vehicle_ID,
  Engine_Size,
  Horsepower,
  Fuel_Efficiency,
  Top_Speed,
  Safety_Rating
)
print(vehicle)
# Install package (Only first time)
install.packages("ggplot2")

library(ggplot2)
```

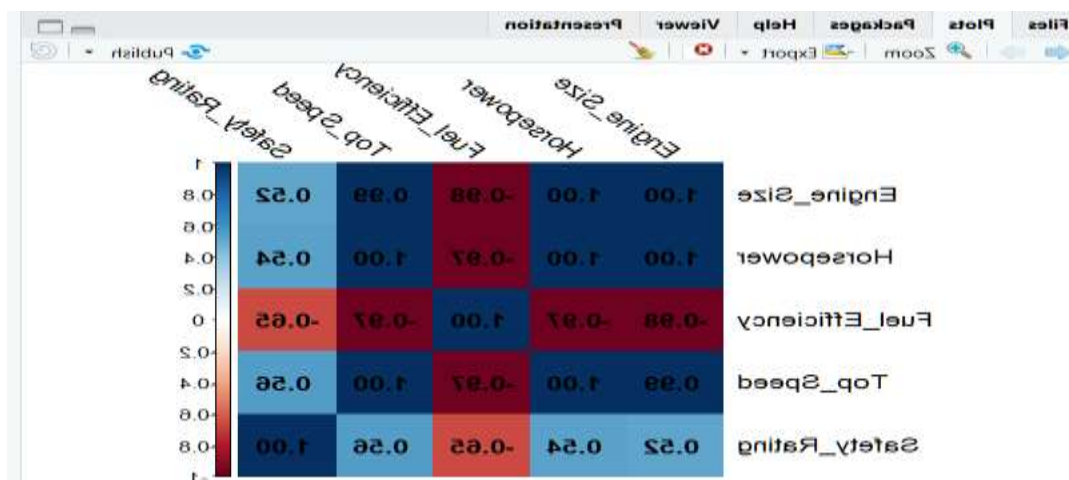


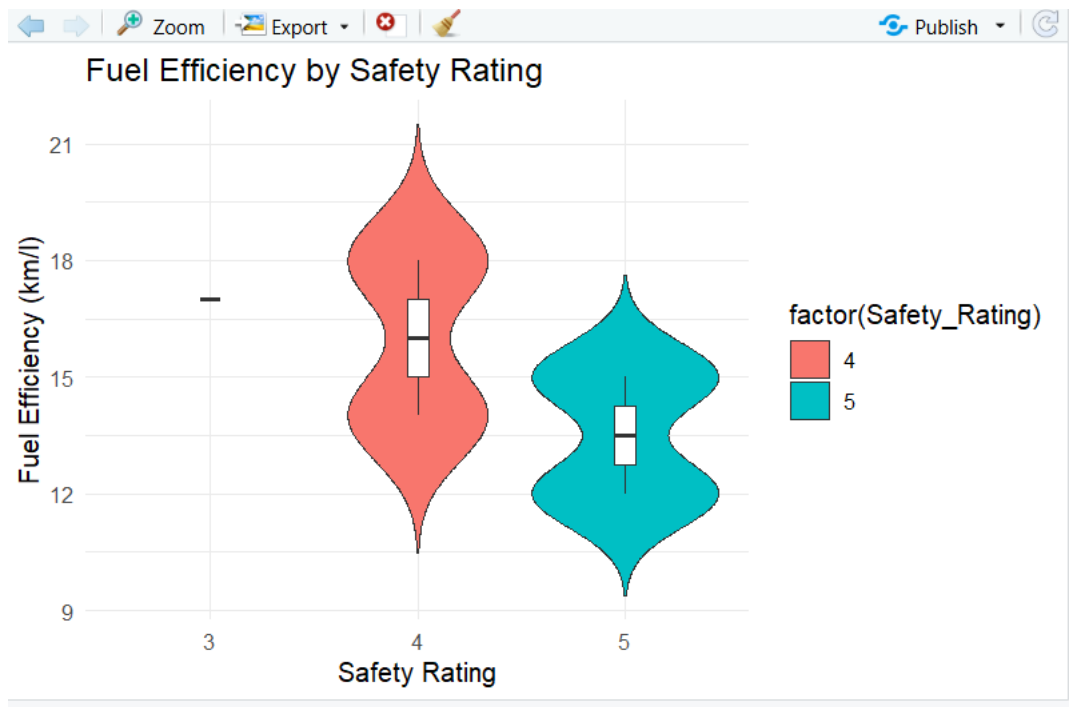
```

ggplot(vehicle,
  aes(x=factor(Safety_Rating),
    y=Fuel_Efficiency,
    fill=factor(Safety_Rating)))+
geom_violin(trim=FALSE)+
geom_boxplot(width=0.1,
  fill="white")+
labs(title="Fuel Efficiency by Safety Rating",
  x="Safety Rating",
  y="Fuel Efficiency (km/l)")+
theme_minimal()
install.packages("corrplot")
library(corrplot)
vehicle_num <- vehicle[,2:6]
cor_matrix <- cor(vehicle_num)
corrplot(cor_matrix,
  method="color",
  addCoef.col="black",
  tl.col="black",
  tl.srt=45)

```

OUTPUT:





18.

Code:

```
# Student Academic Performance Dataset
Student_ID <- c("S1","S2","S3","S4","S5")
Age <- c(19,21,20,22,23)
Study_Hours <- c(12,8,15,10,7)
Attendance <- c(90,70,95,85,60)
Test_Score <- c(85,70,92,80,65)
Participation_Score <- c(8,7,9,8,6)
student <- data.frame(
  Student_ID,
  Age,
  Study_Hours,
  Attendance,
  Test_Score,
  Participation_Score
)
print(student)
```

```
x <- 1:5
```

```
test <- student$Test_Score
```

```
participation <- student$Participation_Score
```

```
plot(x,  
     test + participation,  
     type="n",  
     xaxt="n",  
     xlab="Students",  
     ylab="Scores",  
     main="Test Score and Participation Score")
```

```
axis(1,  
     at=1:5,  
     labels=student$Student_ID)
```

```
polygon(c(x,rev(x)),  
        c(rep(0,5),rev(test)),  
        col="skyblue",  
        border=NA)
```

```
polygon(c(x,rev(x)),  
        c(test,  
          rev(test+participation)),  
        col="orange",  
        border=NA)
```

```
legend("topleft",  
       legend=c("Test Score", "Participation Score"),
```

```

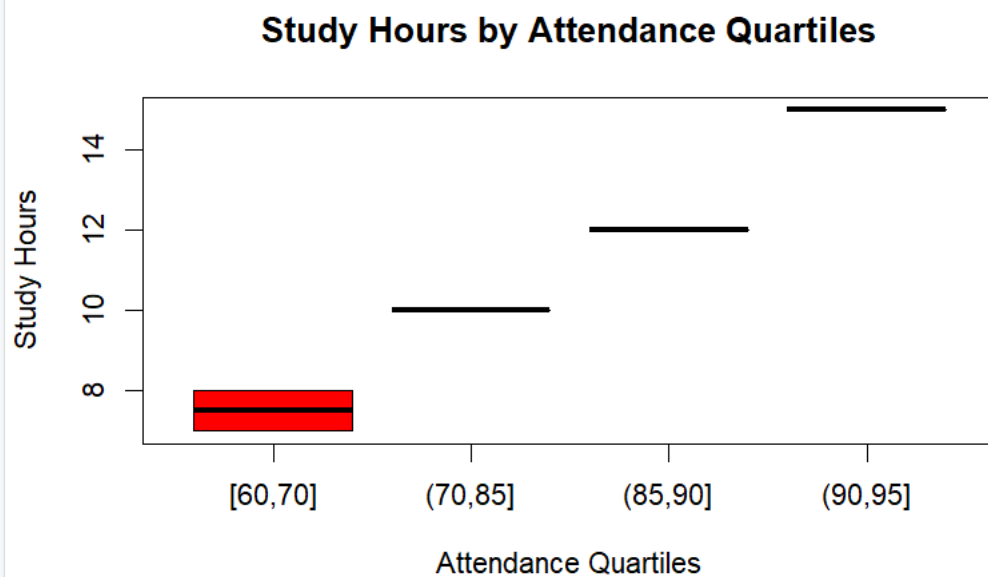
fill=c("skyblue","orange"))
student$Attendance_Group <- cut(student$Attendance,
                                breaks=quantile(student$Attendance,
                                                  probs=seq(0,1,0.25)),
                                include.lowest=TRUE)

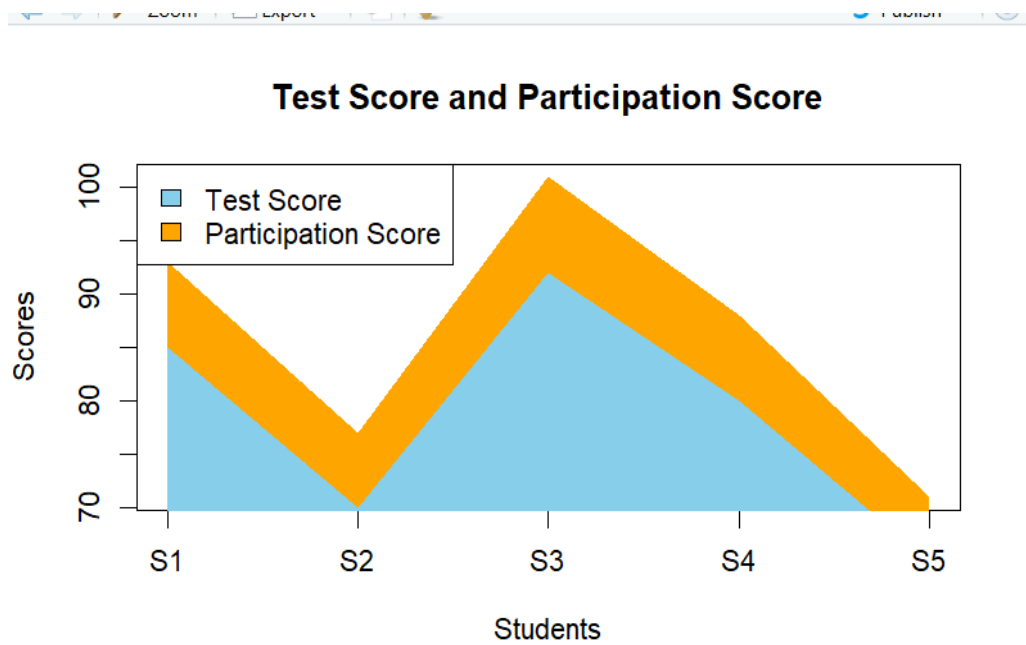
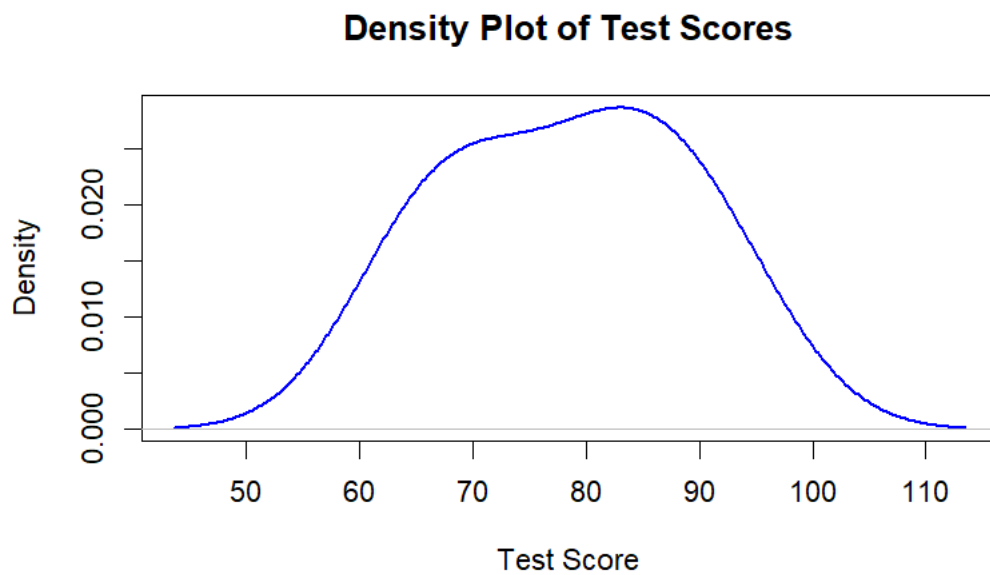
boxplot(Study_Hours ~ Attendance_Group,
        data=student,
        col=rainbow(4),
        main="Study Hours by Attendance Quartiles",
        xlab="Attendance Quartiles",
        ylab="Study Hours")

plot(density(student$Test_Score),
     main="Density Plot of Test Scores",
     xlab="Test Score",
     col="blue",
     lwd=2)

```

OUTPUT:





19.

Code:

Mobile App Usage Dataset

```
User_ID <- c("U01","U02","U03","U04","U05","U06")
```

```
Gender <- c("Male","Female","Male","Female","Male","Female")
```

```
Age <- c(20,22,19,21,23,20)
```

```
Screen_Time <- c(4.5,6.0,3.2,7.1,2.8,5.4)
```

```
App_Usage_Count <- c(18,25,12,30,10,22)
```

```
Data_Used <- c(2.4,3.8,1.6,4.5,1.2,3.1)
Satisfaction <- c(3,5,3,5,2,4)
Usage_Date <- c("2025-01-08","2025-01-08","2025-02-11",
                "2025-02-11","2025-03-14","2025-03-14")
```

```
mobile <- data.frame(
  User_ID,
  Gender,
  Age,
  Screen_Time,
  App_Usage_Count,
  Data_Used,
  Satisfaction,
  Usage_Date
)
```

```
print(mobile)
```

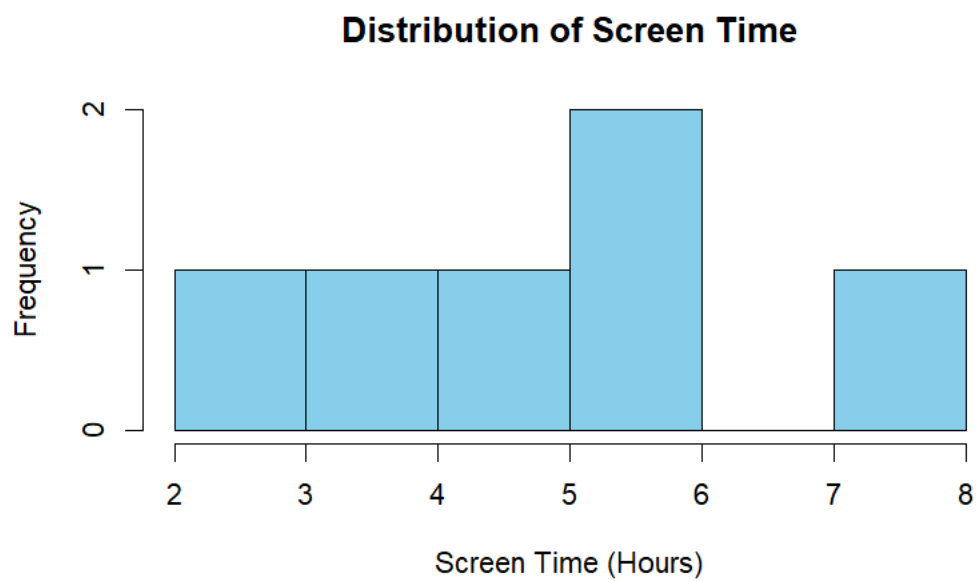
```
hist(mobile$Screen_Time,
     main="Distribution of Screen Time",
     xlab="Screen Time (Hours)",
     ylab="Frequency",
     col="skyblue",
     border="black")
```

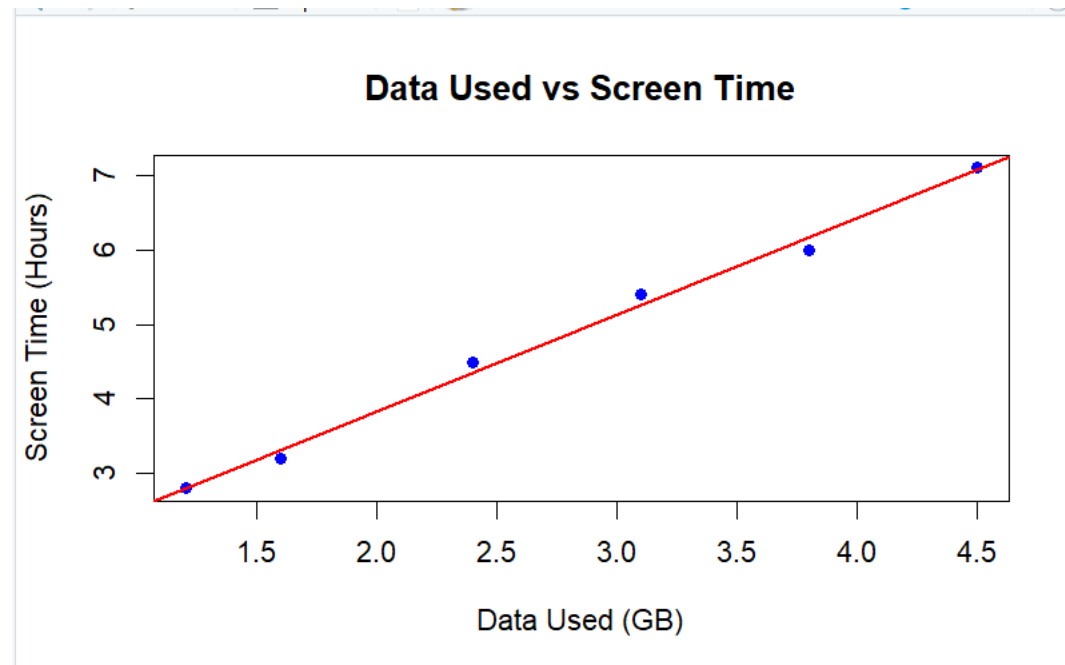
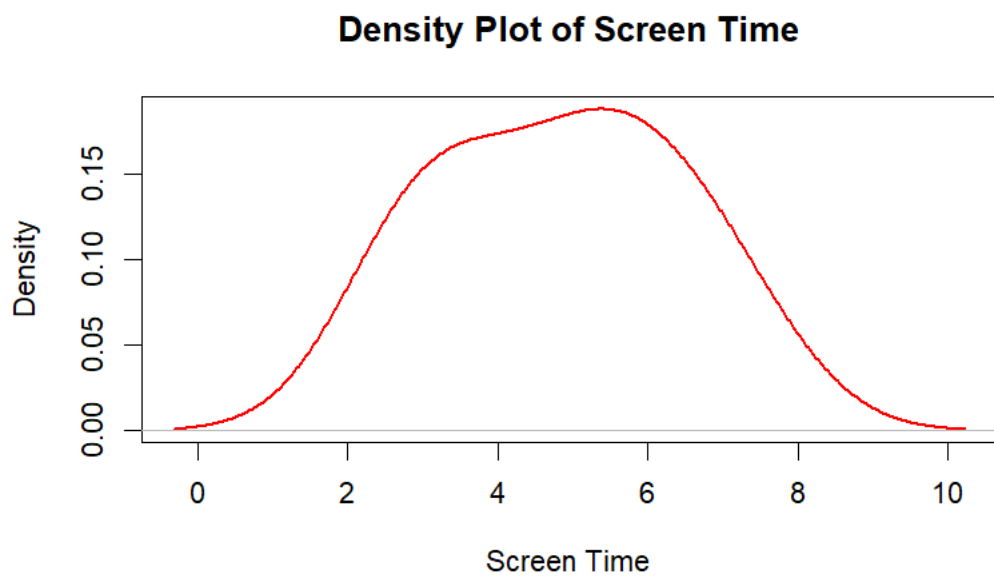
```
plot(density(mobile$Screen_Time),
     main="Density Plot of Screen Time",
     xlab="Screen Time",
     col="red",
     lwd=2)
```

```
plot(mobile$Data_Used,
     mobile$Screen_Time,
```

```
pch=19,  
col="blue",  
xlab="Data Used (GB)",  
ylab="Screen Time (Hours)",  
main="Data Used vs Screen Time")  
  
abline(lm(Screen_Time ~ Data_Used,  
          data=mobile),  
       col="red",  
       lwd=2)  
  
cor(mobile$Data_Used,  
     mobile$Screen_Time)
```

OUTPUT:





20.

Code:

```
# Step 1: Create Dataset
Patient_ID <- c(1,2,3,4,5)
Age <- c(25,40,35,50,29)
Waiting_Time <- c(2,5,1,7,3)
Appointment_Status <- c("Attended",
                        "Missed",
```



```
"Attended",  
"Missed",  
"Attended")
```

```
hospital <- data.frame(  
  Patient_ID,  
  Age,  
  Waiting_Time,  
  Appointment_Status  
)
```

```
print(hospital)
```

```
# =====  
# Task 1 : Histogram of Patient Ages  
# =====
```

```
hist(hospital$Age,  
     main="Patient Age Distribution",  
     xlab="Age",  
     ylab="Frequency",  
     col="skyblue",  
     border="black")
```

```
# =====  
# Task 2 : Pie Chart of Appointment Status  
# =====
```

```
status <- table(hospital$Appointment_Status)
```

```

pie(status,
     main="Appointment Status Distribution",
     labels=names(status),
     col=c("lightgreen","orange"))

# =====

# Task 3 : Bar Chart of Waiting Time

# =====

barplot(hospital$Waiting_Time,
       names.arg=hospital$Patient_ID,
       main="Waiting Time by Patient",
       xlab="Patient ID",
       ylab="Waiting Time (Days)",
       col="pink",
       border="black")

# =====

# Task 4 : Scatter Plot (Age vs Waiting Time)

# =====

plot(hospital$Age,
     hospital$Waiting_Time,
     main="Age vs Waiting Time",
     xlab="Age",
     ylab="Waiting Time (Days)",
     pch=19,
     col="blue")

abline(lm(Waiting_Time ~ Age,

```

```
data=hospital),  
col="red",  
lwd=2)
```

OUTPUT:

